

RSS-Based Access Point Localization Experiment

1 Experimental Setup

This experiment investigates the localization of a single Wi-Fi access point using received signal strength (RSS) measurements collected inside the Faculty of Engineering at Ferdowsi University of Mashhad. The target access point was located in a corridor area of the building. Its independently measured reference coordinate was:

36 deg 18'46.1"N, 59 deg 31'34.9"E

which is equivalent to:

latitude = 36.3128055556, longitude = 59.5263611111.

The building map was drawn from the available JSON map file. In particular, the room and corridor outlines were reconstructed from the polygon coordinates stored in the map data.

2 Data

The initial dataset used in this experiment is an Excel file containing only the measurements associated with the selected target access point:

`selected_ap_b28ba99a8d2a_pivot_data.xlsx`

This file is stored in the output directory of the experiment and contains two sheets:

- **selected_ap_samples**: the measurement records used for localization;
- **summary**: a compact summary of the selected access point and its RSS statistics.

The sample sheet contains the measurement coordinates, device and orientation metadata, and one RSS column for the selected access point. RSS measurements from other access points are not included. The RSS column is named:

`selected_ap_rss_dbm`

The selected access point is:

`BSSID = b28ba99a8d2a`

The valid measurements cover the corridor region around the selected access point and nearby spaces. Figure 1 shows the RSS samples overlaid on the building map. Stronger RSS values are shown using green shades, while weaker RSS values are shown using red shades.

Table 1: Summary of the Excel dataset used in the experiment.

Quantity	Value
Access point BSSID	b28ba99a8d2a
Number of valid RSS samples	1069
Maximum RSS	−29.0 dBm
Mean RSS	−72.89 dBm
Minimum RSS	−98.0 dBm

3 Access Point Reference and Initial Localization Cue

The measured reference coordinate of the target access point was used as a spatial cue for evaluating the selected data. The RSS samples were first summarized by an RSS-weighted centroid. For a sample with RSS value r_i , the weight was defined as:

$$w_i = 10^{(r_i - r_{\max})/10}, \quad (1)$$

where $r_{\max}$ is the strongest RSS value in the dataset. The corresponding weighted centroid was computed as:

$$x_c = \frac{\sum_i w_i x_i}{\sum_i w_i}, \quad y_c = \frac{\sum_i w_i y_i}{\sum_i w_i}. \quad (2)$$

For this dataset, the RSS-weighted centroid was approximately 1.37 m away from the measured reference coordinate. This indicates that the strongest RSS measurements are spatially concentrated around the expected access point location.

4 RSS Fitting Model

The access point location was estimated by fitting a log-distance path-loss model to the RSS measurements. The model is:

$$RSS(d) = RSS_{1m} - 10n \log_{10}(d), \quad (3)$$

where RSS_{1m} denotes the RSS at one meter, n is the path-loss exponent, and d is the Euclidean distance between a measurement point and the estimated access point position.

The unknown parameter vector was:

$$\theta = (x_{AP}, y_{AP}, RSS_{1m}, n). \quad (4)$$

The parameters were estimated by minimizing the residual between the measured RSS values and the RSS values predicted by the model. A robust `soft_l1` loss was used to reduce the effect of outliers, which are expected in indoor RSS measurements due to multipath propagation, shadowing, and device-orientation effects.

5 Localization Results

Table 2 reports the fitted model parameters and the estimated access point location.

Figure 2 shows the reference coordinate and the estimated location on the building map. The red cross marks the reference coordinate, and the black star marks the estimated access point position.

The estimated location was approximately 1.86 m away from the reference coordinate. Given the variability of indoor RSS measurements, this result shows that the log-distance fitting model

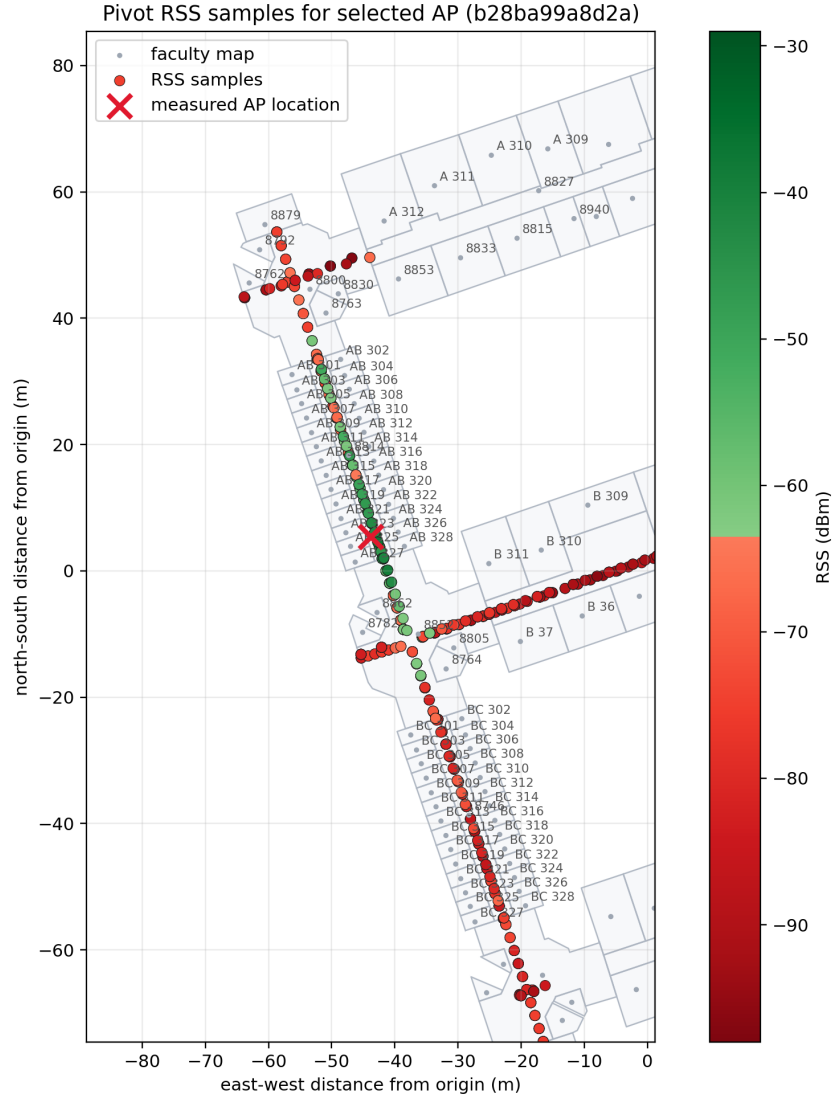


Figure 1: RSS measurements of the selected access point overlaid on the building map.

provides a practical approximation of the access point position when a sufficient number of measurements are available in the surrounding corridor area.

Table 2: Estimated AP location and fitted RSS model parameters.

Quantity	Value
Estimated latitude	36.3128106191
Estimated longitude	59.5263414073
Distance from reference AP	1.86 m
Estimated RSS_{1m}	-28.36 dBm
Path-loss exponent n	3.22
RSS RMSE	8.53 dB
RSS MAE	6.91 dB

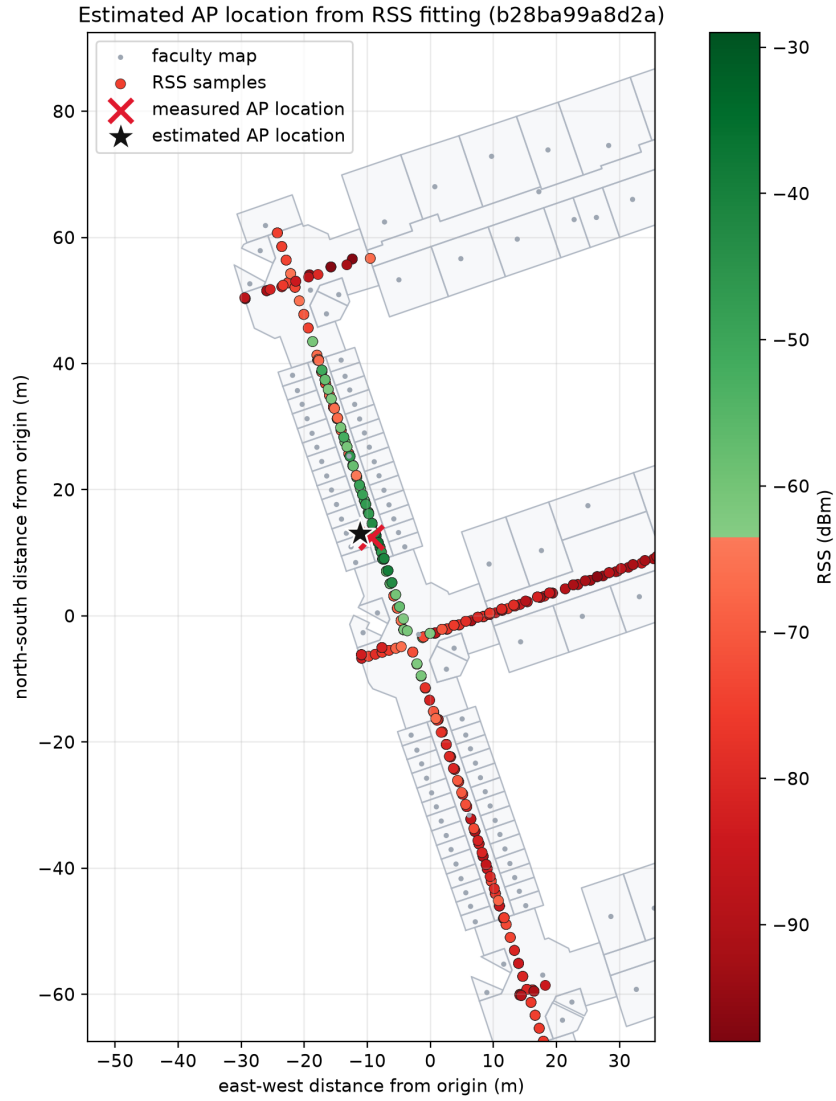


Figure 2: Reference and estimated access point locations on the building map.